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Indoor Air Quality Forecasting With Machine Learning: A Systematic Review

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ABSTRACT

Indoor air pollution poses substantial health risks. People spend over 90% of their time indoors, making the indoor environment a major contributor to overall pollution exposure. Therefore, indoor air quality (IAQ) prediction is critical for proactive mitigation strategies. Despite advances in machine learning (ML), research on IAQ forecasting remains fragmented, with limited integration of occupant behaviour, inconsistent evaluation practices and persistent challenges in predicting high pollution events. This systematic review synthesises evidence from 99 peer-reviewed studies published between 2010 and 2026, examining data sources, modelling approaches, forecasting horizons and performance metrics used for IAQ prediction. The analysis reveals rapid growth in publications after 2020, with deep learning time-series architectures such as long short-term memory (LSTM) dominating because of their ability to capture temporal dependencies, whereas transformer-based models show emerging potential for longer-term forecasting. Only 13% of reviewed studies incorporate behavioural or occupancy-related variables, despite their demonstrated influence on indoor pollution and model accuracy. Model evaluation primarily relies on conventional metrics such as RMSE and R2, with limited assessment of health-relevant thresholds or forecast horizon-specific performance. Geographic disparities are also evident, with developing regions such as Africa and South America under-represented. The review identifies critical research gaps and provides actionable recommendations for integrating behavioural data, adopting context-aware evaluation metrics and addressing data imbalance to improve the robustness and applicability of IAQ forecasting models in real-world buildings.

1 | Introduction

Indoor air quality (IAQ) significantly affects human health, with the World Health Organisation (WHO) attributing approximately 7 million premature deaths annually to air pollution [1]. People spend more than 90% of their time indoors or in vehicles [2], where airborne pollutants such as particulate matter (PM) and volatile organic compounds (VOCs), alongside indoor environmental parameters including carbon dioxide (CO_2), temperature and humidity, can accumulate and influence respiratory, cardiovascular and cognitive health outcomes [3–5]. The WHO reports that 90% of the population is exposed to unhealthy air pollution levels [1], underscoring the urgency of effective IAQ management. This urgency has been reinforced at the policy level, with the United Nations General Assembly recognising clean indoor air as a critical public health issue in its 2025 call for action on IAQ. Meanwhile, the global IAQ monitor market is growing rapidly. In 2024,

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it was valued at 4.25 billion USD and it is projected to exceed 9 billion USD by 2033 [6]. Market analysts claim the biggest market restraints to the cost of sensors, sensor accuracy and the need for regular calibration [6].

Timely IAQ prediction is crucial for improving occupant health. Statistical models such as autoregressive moving average (ARMA), autoregressive integrated moving average (ARIMA) and linear regression were initially used to model IAQ [7]. However, these models suffer poor performance and are unable to predict longer horizons accurately, limiting their effectiveness [7].

Consequently, research attention has shifted towards machine learning (ML) and artificial intelligence (AI) approaches. Advances in these methods, along with the availability of lower-cost sensors, have enabled more accurate time-series forecasting, with techniques such as long short-term memory (LSTM), gated recurrent unit (GRU) and transformers showing promise [8]. However, the literature highlights critical gaps: limited integration of human behavioural data, despite its proven impact on IAQ [9]; difficulty in acquiring data (due to costly and potentially invasive sensors) [7, 10]; and the use of evaluation metrics that do not consider health impact or performance over time. These unresolved challenges have motivated efforts to examine existing research and better understand how IAQ forecasting methods are currently developed and evaluated.

Recent systematic reviews have examined ML applications for IAQ assessment and prediction. For example, Saini et al. (2024) reviewed 18 studies published between 2020 and 2022 with a focus on commonly used algorithms, pollutants and datasets [11]. In contrast, the present work extends this line of research by focusing specifically on forecasting performance and evaluation practices across a broader body of literature (99 studies spanning 2010–2026). Rather than cataloguing methods alone, this study contributes by (1) evaluating reported prediction accuracy across different forecasting horizons; (2) examining the role of behavioural and occupancy-related variables in IAQ forecasting models; and (3) assessing the suitability and consistency of evaluation metrics used to report forecasting performance, providing a critical assessment of how IAQ forecasting models are developed and evaluated in practise.

To evaluate how IAQ forecasting research has developed and how effectively current approaches are assessed in practise, this paper investigates:

- Research question 1: What data are being used for IAQ forecasting?
- Research question 2: What ML methods of IAQ forecasting are well researched?
- Research question 3: How does model accuracy evolve over prediction time?
- Research question 4: What evaluation metrics are suitable for IAQ forecasting?

We systematically review 99 studies published between 2010 and 2026 using the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses), analysing forecasting methods, data use and evaluation practices [12]. The analysis examines forecasting methods, data usage and evaluation practices within IAQ prediction research, with particular attention to how models are assessed and applied in practice. Results show that LSTM-based models currently dominate IAQ forecasting studies but present challenges related to computational efficiency, whereas transformer-based approaches demonstrate emerging potential for long-term forecasting. The review also identifies key methodological gaps: despite evidence that occupant behaviour improves predictive performance, only 13 studies explicitly incorporate behavioural data, and evaluation practices remain largely reliant on conventional metrics that may not reflect real-world health relevance.

This paper provides:

- An overview of IAQ datasets, forecasting approaches and evaluation practices across the literature.
- Evidence-based recommendations for integrating behavioural data and improving geographic diversity in future studies.
- A critical examination of commonly used evaluation metrics, alongside discussion of context-aware alternatives (e.g., threshold-based assessments).

Section 2 establishes the background, detailing key indoor air pollutants and their health impacts, alongside an overview of ML for forecasting. Section 3 details the method of the systematic review. Section 4 presents the results of our systematic review, analysing 99 studies to identify trends in data sources (e.g., sensor networks, third-party datasets), ML methods and the evaluation methods used to measure model performance. Section 5 concludes these findings, discussing practical implications such as the need for standardised evaluation metrics and behavioural data integration and proposing directions for future work.

2 | Background

IAQ forecasting is an interdisciplinary field that combines environmental science with ML to predict future concentrations of key indoor air parameters. This predictive approach focuses on critical variables including temperature, humidity and various pollutants such as CO_2, PM ($PM_{2.5}$ and PM_{10}), VOCs, nitrogen dioxide (NO_2), carbon monoxide (CO) and ozone (O_3). Poor IAQ can negatively impact the health of occupants both in the long term and short term, contributing to mild issues like headaches to severe diseases such as lung cancer and heart disease [3]. Therefore, the incentive to predict IAQ is to allow preventative measures, such as adjusting ventilation and mitigating potential health risks.

2.1 | Key IAQ Pollutants and Sources

There are several dominant pollutants that impact indoor environments. PM, particularly $PM_{2.5}$ and PM_{10}, consists of fine particles with diameters of 2.5 micrometres or smaller and 10 micrometres or smaller, respectively [13]. These particles can originate from indoor activities such as cooking, cleaning and heating, as well as from outdoor pollution infiltration [3, 14].

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Their health implications are well-documented, with strong associations to respiratory and cardiovascular diseases [3, 15]. CO_2, a colourless gas produced through human respiration, accumulates easily in poorly ventilated and highly occupied spaces like offices and classrooms, leading to drowsiness, cognitive impairment and headaches at elevated concentrations [3, 16]. VOCs represent a broad class of chemicals that evaporate at room temperature. These compounds can be human-made or naturally occurring substances, including formaldehyde, benzene, toluene and limonene [10, 17–19]. They are emitted from various household products including paints, aerosols and furniture, have been linked to both short-term irritation and long-term organ damage [18]. NO_2, a reactive gas from combustion sources like gas stoves, exacerbates asthma and reduces lung function particularly among vulnerable groups [3]. Temperature and humidity, although not pollutants, significantly influence comfort levels and affect pollutant dispersion rates and chemical reactions, with high humidity contributing to mould growth [3]. O_3, which forms at ground level through reactions between NO_2, VOCs and sunlight, is known to cause respiratory inflammation [17, 20]. Human activity can have a large impact on air quality, though this factor remains under-represented in many forecasting models as shown later in this paper. CO_2 levels demonstrate a direct correlation with both occupancy and activity intensity, with physical exertion leading to more rapid accumulation [3, 16, 21]. PM concentrations experience sharp increases during specific activities such as cooking, cleaning or burning various fuels [3]. Similarly, volatile organic compound levels surge following the use of aerosol sprays, cleaners and paints [19]. Ventilation behaviours, including window opening and HVAC system usage, significantly alter pollutant dispersion and build up [22].

2.2 | ML forecasting

IAQ forecasting is a complex problem with both spatial and temporal considerations, requiring a large amount of data, specialised time-series ML architectures and additional considerations for evaluation [7, 23]. Model creation begins with data acquisition. This can be from IoT sensor networks that provide high-frequency measurements of target pollutants, building management systems providing details such as ventilation status, temperature and humidity and other sources (e.g. cameras detecting occupancy) [24, 25]. External data streams, including meteorological forecasts and outdoor air quality measurements, can also be incorporated to model the effect of weather and the infiltration of pollutants [9, 13]. These time-series data present challenges, including different sampling frequencies between data sources, missing observations and the need to have a continuous block of temporal data.

IAQ forecasting has grown rapidly in recent years, progressing from classical statistical approaches to modern deep learning architectures. Early systems relied on statistical methods like ARIMA models and linear regression [7, 13]. The introduction of ML techniques such as random forest (RF) and support vector machine (SVM) improved handling of nonlinear relationships and model accuracy [13, 26]. Time-series ML architectures have since been designed, which are specifically adapted for temporal dependencies. Deep learning architectures have proven particularly effective, with LSTMs and GRUs learning temporal patterns through their gated memory mechanisms, whereas transformer-based models with attention mechanisms excel at identifying both local and global temporal relationships [8, 23].

3 | Materials and Methods

PRISMA methodology was used to conduct this systematic review. The PRISMA 2020 checklist has been followed to ensure that the review is transparent and reproducible. This paper is aimed at providing a comprehensive review of the current state of research in IAQ prediction using AI and ML methods. The review focuses on the data and methods used, the features necessary for accurate prediction and the accuracy of current methods. The review aims to answer the research questions, as well as identify areas of potential future research. To address IAQ prediction using AI/ML methods, the following steps have been taken: research questions have been identified, a search strategy has been developed, inclusion and exclusion criteria have been defined, a selection process has been outlined and a bias assessment has been conducted.

3.1 | Search Strategy

Seven databases were searched in June 2026: ACM, IEEE Xplore, ProQuest, PubMed, ScienceDirect, Scopus and Web of Science. The selection of databases was informed by prior evaluations of academic search systems for systematic reviews [27]. The search was undertaken during June 2026. The general search string used was:

- *Title*: (“indoor air quality“ “OR “indoor environment quality“) AND (predict∗ OR forecast∗)
- Keywords and abstract: (“indoor air quality“ or “indoor environment quality“) and (predict∗ or forecast∗) and (“model“ or “algorithm“ or develop∗)

where an asterisk signifies a wildcard. In the event that a database's search engine was not capable of this exact search, the search was modified minimally to run. The exact search strings used for each database can be found in Appendix A (Table A1)

The search strategy intentionally centred on the terms ‘indoor air quality’ and ‘indoor environment quality’ rather than pollutant-specific keywords (e.g., CO_2, $PM_{2.5}$ and VOCs). Preliminary searches using pollutant-level terms returned a large number of studies focused on outdoor air pollution, atmospheric modelling and sensor calibration, which were outside the scope of this review. As this study focuses on IAQ forecasting as a whole rather than prediction of individual pollutants in isolation, the selected terminology was retained to maintain relevance to IAQ-focused research and reduce the inclusion of unrelated studies. Additional terms such as ‘soft-sensor’ and ‘time-series modelling’ were explored during preliminary searches but were not retained in the final query as they reduced the number of relevant results returned.

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The search returned 4992 results before duplicate removal. After removing 2122 duplicates, a remaining 2870 unique papers to process with the inclusion and exclusion criteria.

3.2 | Inclusion and Exclusion Criteria

In order to facilitate the selection process and to ensure that the papers selected were relevant to IAQ prediction using AI/ML methods, papers were only included if they satisfied the following criteria.

Inclusion criteria:

1. Research papers that use AI/ML methods for IAQ forecasting, or forecasting of an IAQ feature.
2. The paper includes details of the dataset used for training in some capacity.
3. At least one meaningful performance metric must be shown.

Exclusion criteria:

1. Duplicate papers.
2. Papers not accessible to the authors.
3. Papers published prior to 2010.
4. Papers where the primary focus is outdoor air quality.
5. Papers where the primary focus is not IAQ forecasting.
6. Papers not written in English.
7. Papers that are secondary studies (such as other review papers).
8. Papers where the indoor environment is a vehicle.

3.3 | Selection Process

Screening was conducted by a single reviewer because of resource constraints. To minimise selection bias, predefined inclusion and exclusion criteria were strictly applied, and uncertain cases were re-evaluated during full-text assessment. The screening process was conducted under the guidance of the wider research team, including academic and industry co-authors, who provided input on the review design and interpretation of eligibility criteria. This limitation is nevertheless acknowledged when interpreting the findings.

The systematic review accelerator toolset by Tera tools was used by the author to facilitate the selection process. More specifically, the duplication removal tool was used to flag potential duplicates for manual review [28, 29]. A PRISMA flow diagram of the paper selection process can be seen in Figure 1.

3.4 | Data Extraction and Analysis

After applying the inclusion and exclusion criteria detailed in Subsection 3.2, 99 papers remained. Guided by consultation with our industry partner, key information such as publication details, AI/ML methods, predicted pollutants, evaluation metrics, datasets and data sources, behavioural variables, geography and study environment was extracted from each study. These variables were coded using predefined rules and analysed descriptively.

Forecast horizon was defined using the reported prediction interval. Where multiple horizons were evaluated, all intervals were recorded and studies were categorised according to the longest reported prediction window to enable comparison across studies.

Behavioural inclusion was defined as the explicit use of occupant-related variables (e.g., occupancy counts, activity indicators or user-driven environmental interactions) as model inputs. Studies relying solely on environmental sensor data were classified as not including behavioural information. More granular categorisation could also consider factors such as the method of behaviour recording (e.g., self-reporting, machine vision and wearable sensors), sensor placement or the spatial context of the monitored environment.

Additional contextual variables, including sensor type, data source, geographic setting and consideration of health impacts or temporal performance, were coded where reported and incorporated into the qualitative synthesis.

ML methods were recorded based on the primary model proposed by the authors, with benchmarking models documented separately. Each publication was treated as a single study during frequency analysis unless model-specific comparisons were required.

Evaluation practices were analysed using reported metrics (e.g., RMSE, MAE, MAPE and R^2). Because pollutants, datasets and measurement units varied substantially across studies, numerical performance values were not normalised; instead, metrics were compared descriptively to identify common assessment practices.

3.5 | Risk of Bias Assessment

Systematic reviews are subject to bias introduced by the review process itself. Potential sources of review-level bias include the choice of search terms and databases, the inclusion and exclusion criteria applied during screening and the fact that screening was conducted by a single reviewer. Although the screening process was undertaken by one researcher, it was carried out under the guidance of the wider research team, which included both academic and industry co-authors, who provided input on the review design and interpretation of findings. These factors may nevertheless have resulted in some relevant studies being missed despite the use of multiple databases and predefined criteria.

In addition, a qualitative risk-of-bias assessment was performed at the study level to support interpretation of findings. Because IAQ forecasting studies are methodologically diverse, studies were examined using ML-relevant domains, including: clarity of training/testing procedures, risk of data leakage, reporting of prediction horizons, transparency of input features (including behavioural variables), adequacy of dataset and sensor reporting, handling of missing data and preprocessing and the suitability of evaluation metrics used.

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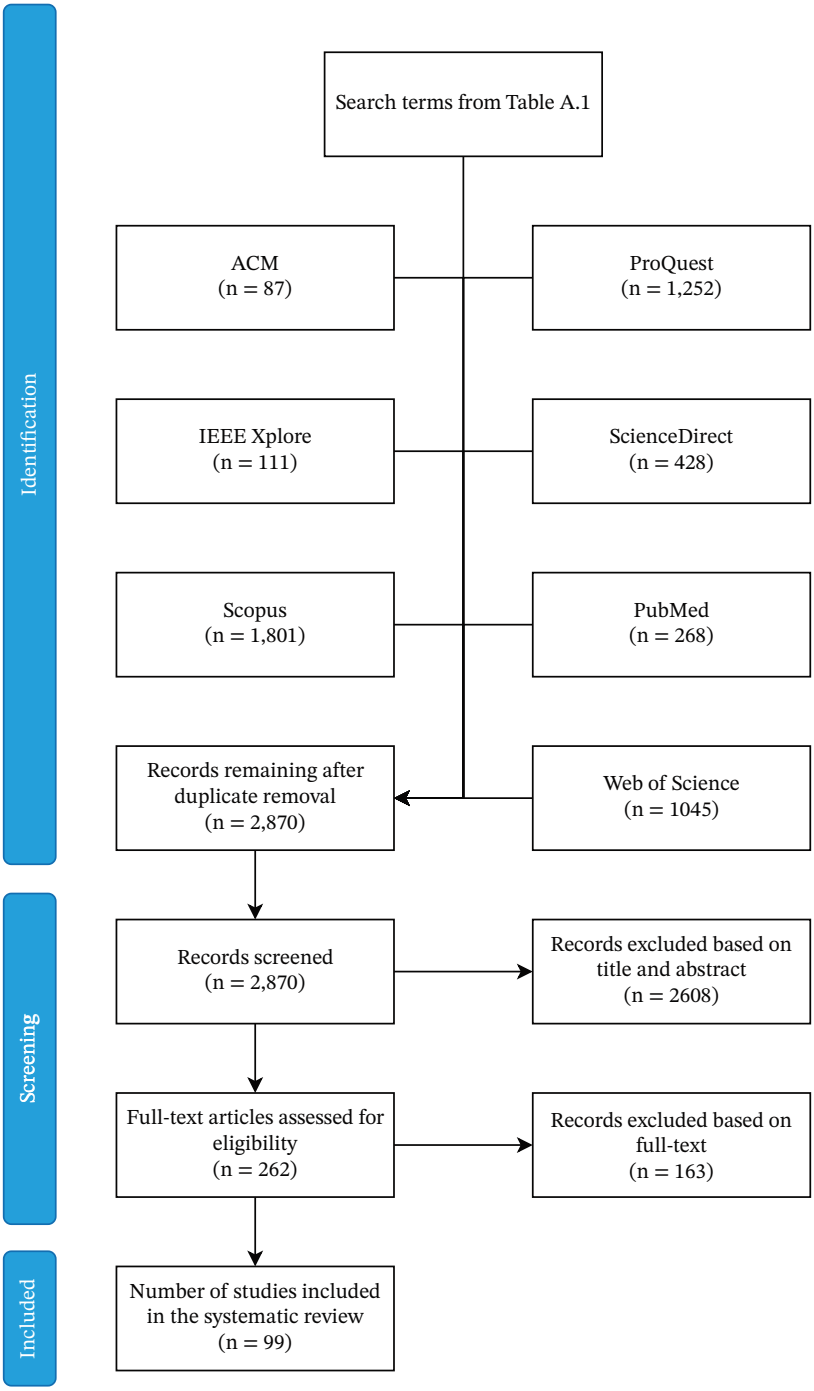


FIGURE 1 | The PRISMA flow diagram for this systematic review.

Risk of bias was judged qualitatively based on completeness of reporting and alignment between modelling approach and evaluation design. Studies were not excluded on this basis; instead, these considerations informed the narrative synthesis and the interpretation of trends reported in Section 4.

3.6 | Registration, Protocol, and Certainty Assessment

This systematic review was not registered, and a review protocol was not prepared. Therefore, no protocol amendments are reported. Reporting bias due to missing results was not formally assessed, and no formal certainty-of-evidence assessment, such as GRADE, was conducted. Given the methodological heterogeneity of IAQ forecasting studies, the findings were synthesised descriptively rather than through meta-analysis.

4 | Systematic Review

IAQ forecasting typically involves data collection, preprocessing, model training and testing. The review aims to provide an overview of the current state of research in each of these areas,

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highlighting the most effective methods and techniques used in IAQ forecasting.

4.1 | Analysis of Publications

Figure 2 highlights the increasing interest in IAQ forecasting, with the number of publications rapidly increasing in the last 5 years compared with the previous decade.

Figure 3 shows that IAQ forecasting studies are concentrated in China, South Korea, the USA, India and Western Europe, with no reviewed studies from Africa, South America or the Middle East. Because of poor air quality disproportionately affecting developing nations, the consequence of this is that the communities that are most affected by poor air quality are not being considered in the literature. This is made worse by the fact that the environment and climate has a significant impact on IAQ; models developed in one region may not be generalisable to another for this reason.

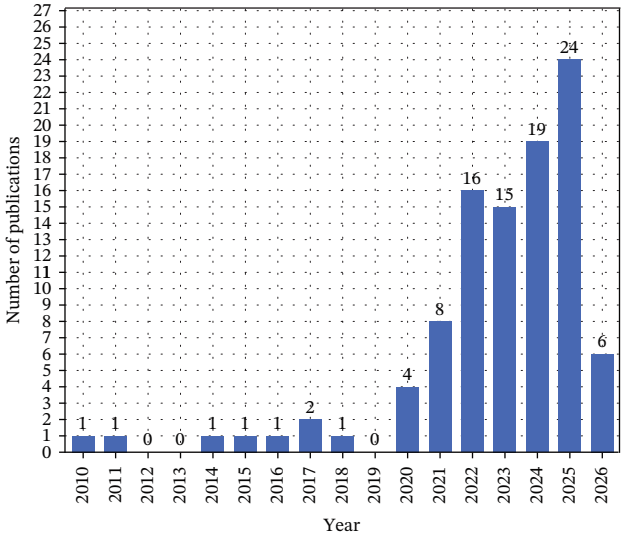


FIGURE 2 | The number of publications per year (2010–2026).

Figure 4 shows that educational, residential, subway and office spaces dominate the literature, likely reflecting easier researcher access. Hospitals and shopping malls remain under-studied despite their high occupancy and health relevance. It is positive that residential, work and educational spaces are well researched as these are locations that people spend the majority of their time. The lack of research in hospitals and shopping malls is troubling as these are people-dense locations, and for hospitals health outcomes can be impacted by air quality [30]. Kim et al. provide a recent example of IAQ forecasting in hospital environments, using IoT sensors to collect long-term data from general wards and negative pressure isolation wards [31]. Their study demonstrates the value of continuous monitoring in healthcare spaces, where IAQ is affected by patient activity, medical procedures, HVAC operation and outdoor air conditions.

Finally, Figure 5 shows that the most common target variables are $PM_{2.5}$, CO_2, PM10, temperature, humidity, NO_2 and VOCs. The emphasis on $PM_{2.5}$ and CO_2 is likely due to the health impacts of these pollutants, as well as their prevalence in indoor environments. Human behaviour generates excess CO_2, which can build up to dangerous levels if the space is not properly ventilated. $PM_{2.5}$ is a common pollutant in indoor environments, generated through activities such as cooking, cleaning and heating, as well as being introduced from outdoors.

4.2 | Research Question 1: What Data Are Being Used for IAQ Forecasting?

Collecting and preparing data is one of the most crucial steps in developing any ML model, and IAQ forecasting is no exception. Poorly prepared data can lead models to learn incorrect patterns, resulting in poor performance and inaccurate predictions. In this section, we explore the data used in the papers reviewed, including the types of data collected and the sensors used, with particular emphasis on how human behavioural data are incorporated into existing research and models. We discuss the challenges and

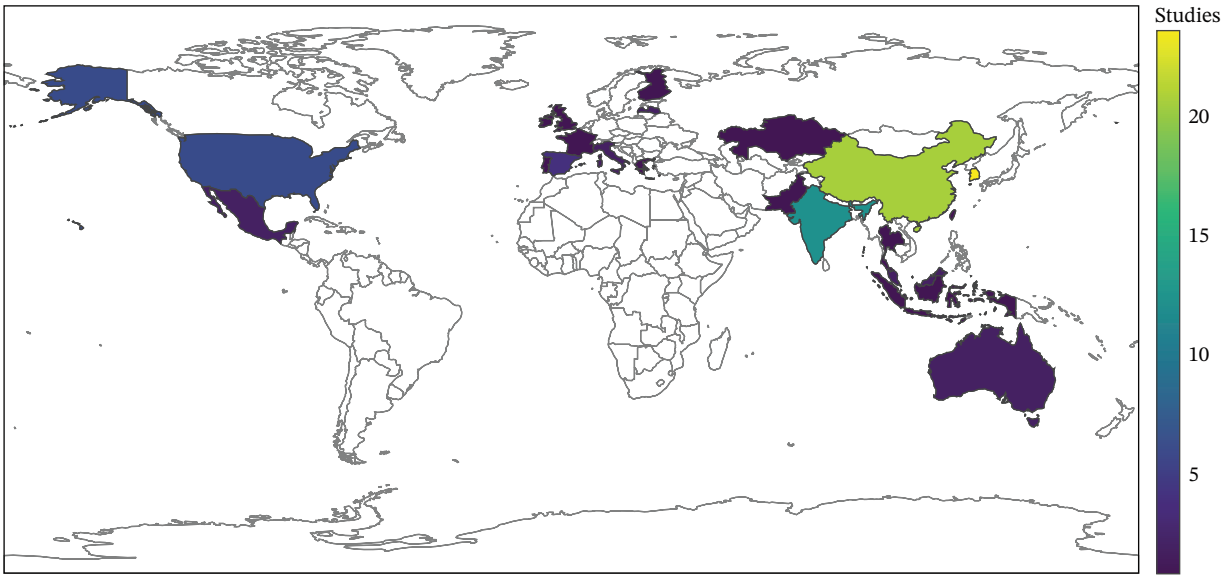


FIGURE 3 | A map of the distribution showing where the reviewed studies were conducted.

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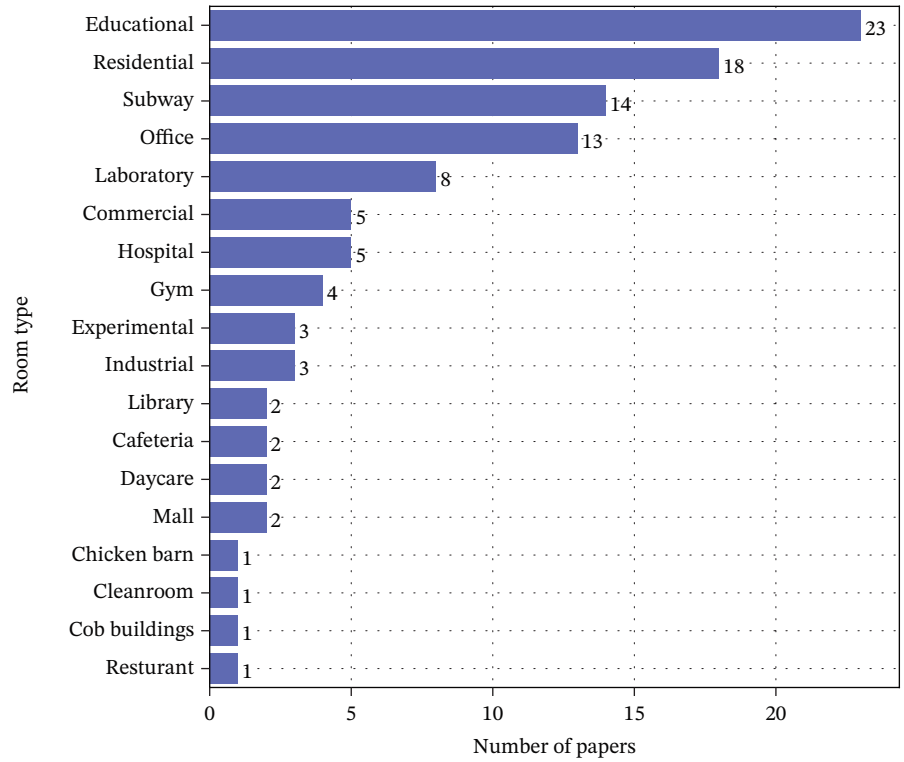


FIGURE 4 | The frequency of different room types in which the studies were conducted.

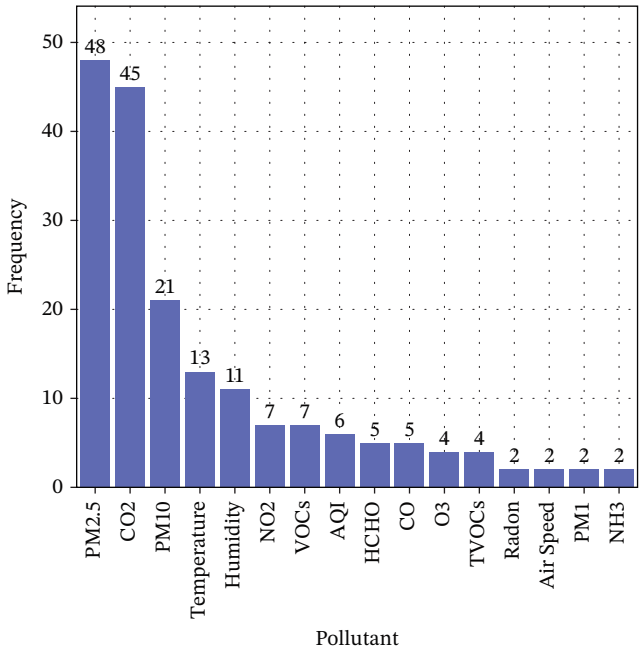


FIGURE 5 | Frequency of target variables forecasted in the literature.

limitations of the data used in IAQ forecasting and suggest potential areas for future research.

4.2.1 | Data Collection

When collecting data, most papers designed their own systems, often containing several commercially available sensors attached to some small computer (e.g. Raspberry Pi and smartphone). Many papers would allow for their sensors to be viewed via IoT dashboards, allowing for a real-time monitoring of the data and the ability to investigate ways of presenting this data to occupants [31, 32].

4.2.1.1 | Sensors. Sensor choice involves trade-offs between cost, accuracy, usability and availability. Throughout the papers reviewed there is no overall trend in the sensor models used, with a total of 135 different sensors being used across the 99 papers reviewed. By filtering out sensors that were only used once, Table 1 shows that for measuring PM, the most common sensors were the Plantower PMS5003 and PMS7003, Nova Fitness SDS011 and the ChinaWay CW-HAT200S. Among these sensors, the ChinaWay has the better accuracy, followed by the Plantower sensors. However, the resolutions and accuracies are not hugely different between sensors and the minimum sample rate is typically similar at around 1 Hz, with the exception being the CW-HAT200S sampling once per min. For measuring temperature and humidity the Aosong Electronics DHT22 and DHT11 are very popular devices made especially for the task with accuracy and good resolution. Bosch's BME680 sensor may also be a convenient option as it measures IAQ Index, VOCs and pressure on top of temperature and humidity, allowing for an easier experimental setup if those features are required or alternatively the Sensiron SCD30 would be a good option in the case that the study was centred around CO_2.

4.2.1.2 | Human Activity Data. Recent work by Godasiaei et al. (2026) highlights the importance of integrating socio-economic, behavioural and environmental variables when modelling household air pollution [56]. Using survey data from 1200

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TABLE 1 | The most frequently used sensors among the reviewed papers.

Sensor	Senses	Accuracy (±)	Resolution	Data sheet	Papers
BME680	IAQ Index, Temperature, humidity, pressure, VOCs	*Humidity* = ±3 %, *VOCs* = ±5%, *pressure* = ±0.6 hPa, *temperature* = ±1°C	*IAQ Index* = 1, *Humidity* = 0.008 %, *VOCs* = 0.08%, *pressure* = 0.18 hPa, *temperature* = 0.1°C	[33]	[24, 34, 35]
SDS011	$PM_{2.5}$, PM_{10}	±15%, ±10 $\mu g/m^3$	0.3 $\mu g/m^3$	[36]	[37–39]
DHT22	Temperature, humidity	*Temperature* = ±0.5 °C, *humidity* = ±2%	*Temperature* = 0.1 °C, *humidity* = 0.1%	[40]	[39, 41, 42]
PMS5003	$PM_{2.5}$, PM_{10}	±10% @ 100–500 $\mu g/m^3$, ±10 $\mu g/m^3$ @ 0–100 $\mu g/m^3$	1 $\mu g/m^3$	[43]	[24, 39, 41]
CCS811	TVOCs, eCO_2	—	—	[44]	[37, 38]
CW-HAT200S	$PM_{2.5}$, PM_{10}	±5%	0.001 mg	[45]	[46, 47]
DHT11	Temperature, humidity	*Temperature* = ±2 °C, *humidity* = ±5%	16 bit	[48]	[37, 38, 49, 50]
MQ135	NH3, NOx, alcohol, benzene, smoke, CO2	—	—	[51]	[42, 49, 50, 52]
PMS7003	PM_1, $PM_{2.5}$, PM_{10}	±10% @ 100–500 $\mu g/m^3$, ±10 $\mu g/m^3$ @ 0–100 $\mu g/m^3$	1 $\mu g/m^3$	[53]	[22, 54]
SCD30	CO_2, temperature, humidity	CO_2 = ±30 *ppm* ±3 % MV, *Temperature* = ±0.4 °C, *Humidity* = ±3%	CO_2 = 1 ppm, *Temperature* = 0.1 °C, *Humidity* = 0.1%	[55]	[24, 35]
BH1-A3	PM_1, $PM_{2.5}$, PM_{10}, PM_{25}, PM_{50}, PM_{100}, CO_2, HCHO, VOCs	*PM* = ±15%, CO_2 = ±40 ppm, *HCHO* = ±0.02 mg/m^3, *VOCs* = ±0.1 mg/m^3	—	—	[46, 47]

households, field measurements, stove-emission testing and ML models, the study showed that stove type, cooking practices, fuel choice and environmental conditions strongly influence pollutant concentrations such as PM, CO and NO_2. This supports the need for IAQ forecasting studies to move beyond sensor-only datasets and incorporate contextual household and behavioural factors. Only 13 of the 99 reviewed papers incorporated human behaviour into their IAQ prediction models data. A summary of this can be seen in Table 2.

The most common method of achieving this has been to incorporate occupancy level into the model [57, 63]. Occupancy is a promising feature since even without specific actions being considered, each person in a room has an individual effect on pollutant build up, so there is likely to be value in using this data. An additional bonus is that for time-series models, the occupancy data can easily be incorporated as a time series feature. To do this there are a few methods; three papers use machine vision techniques to good effect [24, 58, 59]. Kim et al., 2024 ask participants to log their occupancy levels as well as the window, door, AC and air purifier states for usage in their models [22]. An interesting easy option is presented by Cho et al., where because the room being a classroom, the number of occupants can be fairly reliably estimated; however, they do not evaluate the impact of including this data in their model [60]. The remaining papers do not explicitly state how they acquired occupancy data. Papers that evaluate the impact of occupancy have reported a positive impact. Zhou et al. conducted a comparison without occupancy which revealed a clear negative impact in $PM_{2.5}$ prediction when occupancy was removed [58]. Hindriyandhito et al. noted that

TABLE 2 | Summary of human behaviour related features and publications.

Feature	Count	Publications
Occupancy	9	[21, 22, 24, 57–62]
Window state	4	[22, 57, 63, 64]
Door state	3	[22, 57, 63]
Motion	2	[24, 65]
Air purifier	1	[22]
Air conditioner	1	[22]
Physical activity	1	[16]

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despite the occupancy data having a lower correlation coefficient than other features, the CO_2 prediction results showed it had a greater effect on the prediction outcome [59].

Motion has been used in a couple of papers [24, 65]. There are a few reasons why motion might be a good feature to include; for example, it can be an indicator of occupancy levels and it can demonstrate the activity level of the occupants (which might influence pollutants such as CO_2 and PM among others). In a study by Kallio et al., the aggregated motion over a minute was recorded and used as a feature [65]. The authors found that the motion data improved the model results, particularly on predictions longer than a minute.

Two studies used wearable sensorised t-shirts to incorporate occupants' physical activity into IAQ forecasting models [16, 66]. Rescio et al. [16] compared 1D-CNN, RNN and LSTM models trained with and without physical activity data, finding that models including activity data achieved lower RMSE values, with reductions of 5.17, 13.23 and 3.50 for the 1D-CNN, RNN and LSTM models, respectively. Leone et al. [66] extended this approach by combining a Smartex WWS t-shirt with an environmental sensor to forecast CO_2 using Generative adversarial network-based recurrent models. Their results showed that including activity level substantially improved 120-min CO_2 forecasting, reducing RMSE from 80.22 to 7.78 ppm for the LSTM model and from 122.28 to 52.41 ppm for the GRU model. These findings suggest that physical activity is a valuable behavioural feature for IAQ prediction, particularly for CO_2 forecasting, although the authors note that less intrusive devices such as smartwatches may be preferable for real-world deployment.

Kim et al., 2024 incorporate occupancy, air purifier state, window state, AC state and door state [22]. These are some interesting features to include, as each can have a significant impact on the air quality of a room. For example, an air purifier may lower PM levels, whereas an open window may increase PM levels (or tend them towards the outdoor state). The authors noted that airflow was much higher when both the door and window were open compared with just the window, and that the air purifier had a significant impact on $PM_{2.5}$ levels.

These findings indicate that incorporating human behaviour into IAQ prediction models improves the accuracy of the resulting models. Additionally, the use of machine vision and cheap cameras makes detecting occupancy levels a relatively easy task, provided there is a suitable location for the camera and the occupants are comfortable with being recorded. There is still a lot of potential for further research in this area. For example, there are several household tasks known to have a large impact on IAQ such as cooking, cleaning and forms of heating (e.g. burning wood, gas or coal) that are not considered in the papers reviewed. Incorporating human behaviour into IAQ prediction models can be a considerable challenge, collecting the data can be troublesome as it either requires the occupants to report their behaviour (potentially leading to inaccurate data, or difficulty recruiting participants) or requires more sensor setup. Potential solutions to this problem could be to use machine vision techniques to detect human behaviour; additionally, IoT enabled devices are becoming increasingly common and may offer cheap and easy ways to collect data on human behaviour. For example, smart plugs can be used to detect when a device is in use and smart thermostats can be used to detect when a heating system is active.

TABLE 3 | Summary of international third-party datasets used in IAQ prediction.

Third party dataset	Number	Publications
GAMS indoor air quality Dataset	4	[26, 68–70]
European Centre for Medium and Long-Range Weather Forecasts	4	[9, 14, 64, 71]
AirNow	2	[72, 73]
PurpleAir Network	1	[72]
OpenWeather	1	[34]
OpenStreetMap	1	[13]
BEACO2N	1	[72]
DALTON	1	[74]
MERRA-2	1	[64]

4.2.1.3 | Third-Party Datasets. Twenty-eight third-party datasets were used, mainly to add meteorological or outdoor air quality data [64, 67]. As a result, many of the third-party datasets used are very specific to the study's geographical location and provided by regional authorities. By removing the local datasets, we can see the resulting international datasets in Table 3.

- GAMS IAQ Dataset: This dataset is publicly available and contains various indoor and outdoor air quality features [75]. This includes $PM_{2.5}$, PM_{10}, CO_2, humidity, temperature and VOCs. It also contains outdoor features such as CO, NO_2, O_3 1 h average, O_3 8 h average, $PM_{2.5}$, PM_{10} and SO_2. The data is sampled minutely, and the dataset contains 135,100 records taken between November 21, 2016 and March 28 2017. The data was collected in Jing'an, Shanghai.
- European Centre for Medium and Long-Range Weather Forecasts (ECMWF): The ECMWF produces weather forecasts and conducts climate research while also providing a wide range of publicly accessible global meteorological data [76]. One of the most popular datasets they provide is the ECMWF Reanalysis v5 (ERA5), covering January 1940 to present and providing hourly estimates of a range of atmospheric, land and ocean features as well as the uncertainties of the values [77].
- EPA's AirNow: The EPA's AirNow is a United States government service that provides air quality data from the United States, Canada and Mexico [78]. Data can be downloaded directly via pregenerated files or via their API. Data contains site descriptions, monitor descriptions, annual summaries, daily summaries, hourly data and 8-h average data and is sourced from the EPA's air quality system [79, 80].

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- PurpleAir Network: This dataset includes a network of low-cost community-run air quality sensor measurements that can be accessed via the PurpleAir Network API [81]. PurpleAir claim to be the largest public air quality dataset with 35,000 sensors operating worldwide. You can filter between indoor and outdoor sensors and pollutants. Since these sensors are community-run, it is important to note that the placement and operational status of the sensors can vary greatly. To this end PurpleAir assign confidence scores to sensors readings and some level of error detection [82].
- OpenWeather: OpenWeather is an API that provides weather data and forecasts around the world [83]. The API provides current weather and forecasts, government weather alerts, 46-years of historical data. Features available include temperature, pressure, humidity, wind, rain, among others at minutely, hourly and daily level forecasts. The API is free to use for up to 1000 calls a day at which point you must pay per call.
- OpenStreetMap: This is an open source, community driven mapping project that provides a wide range of geographical data. It is limited by the data that the community has collected, but entirely free to use. In terms of air quality prediction, [13] used OpenStreetMap to obtain the geographical location of potential emission sources and incorporate them into their research.
- $BEACO_2N$: The Berkeley Environmental Air-quality and CO_2 Network ($BEACO_2N$) is a low-cost, high-density sensor network designed to provide near real-time street-level measurements of greenhouse gases and air quality [84]. The network is primarily located in the San Francisco Bay Area and provides measurements such as CO_2, CO, NO, NO_2, O_3, and PM. Data can be downloaded directly from the $BEACO_2N$ network map by selecting monitoring locations, date ranges and variables. Since the network uses low-cost sensors, calibration and sensor drift correction are important considerations when using the dataset for air quality prediction.
- DALTON: DALTON is a large-scale IAQ dataset collected using distributed air quality monitors across 30 indoor sites in India [85]. The dataset contains measurements from residential households, studio apartments, food canteens, classrooms and research laboratories across rural, suburban and urban regions. It includes indoor temperature, humidity and several pollutants, including CO2, VOCs, PM1, PM2.5, PM10, NO2, C2H5OH and CO. DALTON also includes activity annotations, making it particularly useful for studying the relationship between occupant behaviour, daily activities and indoor pollution levels.
- Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2): MERRA-2 is a global atmospheric reanalysis dataset produced by NASA's Global Modeling and Assimilation Office (GMAO) [86]. It provides meteorological and atmospheric data from 1980 onwards, with variables such as air temperature, wind, pressure, humidity, aerosols and other atmospheric fields. Although MERRA-2 is not an indoor dataset, it can provide external environmental features that influence IAQ, such as temperature, humidity and aerosol loading.

4.2.1.4 | Imbalanced Data Condition and Augmentation. A challenge in IAQ prediction is that the retrieved data is often highly unbalanced, meaning that the majority of recordings fall within a narrow range. For instance, $PM_{2.5}$ levels may only exceed dangerous thresholds occasionally during specific activities. This imbalance can bias models towards the majority range, reducing performance on the minority group, which is often the most critical to predict as it represents hazardous pollutant levels. To address this issue, one option is to generate synthetic data to balance the dataset. While such approaches are common in classification tasks, applying them to time-series forecasting is more complex, as the generated data must preserve temporal dependencies and respect the continuous nature of the target variable. Existing IAQ studies have rarely explored this direction; however, recent work has begun to address this gap. Asaei-Moamam et al. proposed the DAerosol.GAN.NTM framework, which integrates a neural Turing machine with a GAN to generate synthetic data and replace missing or corrupted records. Their results showed substantial improvements in accuracy, precision and error reduction for $PM_{2.5}$, PM_{10} and AQI forecasting compared with conventional deep learning models [87]. Similarly, another study explored data-driven augmentation methods for air quality prediction, highlighting how synthetic data generation can enhance the reliability of forecasts under sparse or imbalanced conditions [88]. Additionally, other papers have mentioned the possibility and intention of developing GAN methods for IAQ data generation [72, 89]. Beyond GANs, augmentation techniques such as SMOTE and its temporal adaptations (e.g., T-SMOTE), as well as advanced generative models like TimeGAN and T-CGAN, also offer promising avenues for enriching datasets and mitigating imbalance in time-series air quality prediction [90–92].

4.2.2 | Data Preprocessing and Feature Engineering

Data preparation is an integral step to the creation of any ML model. By reviewing the papers we can see that there are four key steps that must be taken before any training begins: (1) data cleaning, (2) feature engineering, (3) preprocessing and (4) data-splitting.

Data cleaning is the process of correcting problems in the collected data. The first step is to identify any outliers using statistical methods and remove them from the dataset [7]. This is important as if a sensor experiences a malfunction, it can produce inaccurate values that could have a negative impact on the model's performance. After this step, missing values should be dealt with [93]. Missing values can originate from several sources such as sensor downtime, communication error or human error as well as from outlier removal. To remedy these gaps there are several different methods that can be used, such as interpolation, forward filling, backward filling or even using a model to predict the missing values [89, 93]. The choice of method will depend on the data and the context of the problem, for instance, a simple method like linear interpolation will work fine for small gaps in the data, but for long gaps (potentially days in some instances), this technique could result in data that does not represent the real world. If the sensor readings are not aligned resampling may also be needed [31, 65].

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Feature engineering is creating new features from the existing data. This can be done by combining features, transforming features or creating new features based on domain knowledge. In the case of IAQ forecasting, a common feature engineering technique is to create a lagged version of the target variable, which can help the model learn temporal patterns in the data [10, 23, 57, 94]. For example, if the target variable is $PM_{2.5}$, a lagged version of this feature could be created by taking the $PM_{2.5}$ value from a previous timestep. Another option is to take an average of values from several previous timesteps, e.g., a 6-h average [22].

Time and date are also useful features, allowing models to learn seasonality. Timestamps can be broken down into year, month, day, h, min, and s depending on data granularity. For example, a 'day of the week' feature might be represented by digits 1–7 to capture weekly trends (e.g., weekends) [62, 72]. Features indicating holidays or seasonal patterns can also be created, and transformations such as sine and cosine of the day or month can help models capture cyclical trends [57].

Beyond time-based features, other contextual features can significantly enhance model performance. These include climatic conditions (temperature, humidity and wind speed), location information (geographical coordinates, altitude and proximity to traffic), space and area characteristics (room size, floor level and ventilation), human activities (occupancy levels and usage patterns) and building materials (walls, flooring and insulation) [9, 58, 59]. These features provide important context that can influence IAQ and therefore impact a model's predictive ability.

Preprocessing is the process of finalising the data before it is fed into the model. This usually includes normalisation—the process of scaling the data to a specific range (e.g. 0–1) [16, 22]. This is an important step as it prevents features with larger values from dominating the model's learning process. During the preprocessing stage data augmentation techniques can be used to address limitations datasets, such as class imbalance, sparsity or limited labelled samples. In other fields, oversampling techniques like synthetic minority oversampling technique (SMOTE) have been used to generate synthetic samples of minority classes, reducing bias and improving prediction accuracy. Likewise, more advanced generative approaches, such as GANs and conditional generative adversarial networks (CGANs), have been applied to synthesise realistic sensor data, enriching datasets and improving model robustness. As mentioned previously, these techniques are not common in the IAQ domain yet. This is possibly because they are harder to apply to the time-series data. Despite this, there are some time-series variations of these data-generation methods such as temporal-oriented synthetic minority oversampling technique (T-SMOTE), TimeGAN and T-CGAN and some recent IAQ specific efforts [87, 88, 90–92].

Data-splitting. Finally, the data should be split into at least training and testing sets, with a validation set also being a good idea. Typical practices for a data split is usually somewhere around 70%–80% for training and the remaining split between validation and testing [16, 95]. If the number of samples is too large to train a model because of computation constraints, a slice of the data can be chosen. In this event considerations should be made to maintain as much seasonality data as possible (e.g. by keeping a whole year/cycle). It is important to split the data in a way that preserves the chronological order of the data (e.g. if you have 100 days of data, you might use the first 70 days for training, the next 15 days for validation and the final 15 days for testing).

4.2.3 | Feature Relevance

Selecting the correct features is crucial for the success of IAQ prediction models. The features used can have a significant impact on the model's performance, and selecting the wrong features can lead to poor results. In this section, we explore the most commonly used features in IAQ prediction models, their relevance to the target variables and how they can be used to improve the model's performance.

$PM_{2.5}$ shows strong positive correlations with PM_{10} and CO_2, as well as moderate correlations with NO_2 and NO [89]. Outdoor $PM_{2.5}$ and PM_{10} levels are consistently identified as good predictors of indoor concentrations [39, 63, 72, 96]. Temperature and humidity also play important roles, though their effects vary; temperature and $PM_{2.5}$ are usually positively correlated, whereas humidity tends to have an inverse relationship [9, 41]. Additionally, contextual factors such as occupancy, ventilation and proximity to pollution sources have been shown to impact $PM_{2.5}$ levels [9, 97].

PM_{10} shares many similarities with $PM_{2.5}$. $PM_{2.5}$ and CO_2 have the strongest positive correlations with PM_{10} [89]. It is important to note that PM_{10} measurements can contain $PM_{2.5}$ and authors should take this into consideration when modelling to alleviate issues such as over fitting. A solution to this issue may be to subtract the $PM_{2.5}$ value from the PM_{10} value, to create a $PM_{10-2.5}$ value while keeping the original $PM_{2.5}$. Similar to $PM_{2.5}$, outdoor PM levels are associated with indoor levels [39, 72]. Temperature and humidity again display associations, with temperature often positively correlated with PM_{10}, whereas higher humidity tends to show a inverse relationship [38, 41]. PM_{10} is also related to gaseous pollutants like NO_2 and CO [89]. Like $PM_{2.5}$, occupancy is also a consideration [97]. Time of day and weak of year are also associated with PM_{10} levels [72].

CO_2 levels have strong relationships with both human activity and environmental factors. Occupancy consistently emerges as an important predictor, with studies reporting positive correlations with human-centred features such as occupancy and physical activity [16, 59, 62]. CO_2 is commonly correlated with PM ($PM_{2.5}$ and PM_{10}) and other gaseous pollutants (NO, NO_2, and CO) [89, 97]. Temperature displays moderate to strong positive correlations with CO_2 across multiple studies, whereas humidity shows weaker and more variable relationships [68, 98]. Temporal factors, such as time of day and weekday versus weekend variations, also prove important [72].

Temperature exhibits moderate to strong positive correlations with CO_2 [59, 98]. As it might be assumed, indoor temperature can be significantly impacted by outdoor temperature [41]. Across the papers reviewed, there is generally a negative correlation between temperature and humidity with the magnitude varying [89, 98]. Associations with pollutants show more inconsistency, with NO_2 displaying both positive and negative correlations in different contexts [39, 89], whereas $PM_{2.5-10}$ generally shows weak to moderate positive relationships [39]. Varying

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correlations may indicate that temperature's relationship with pollutants is context dependent. Finally, there is also a positive correlation with occupancy [59].

Humidity displays many relationships with other features within the reviewed papers, in particular it has a negative correlation with temperature [41, 89, 98]. As with other variables, indoor humidity is greatly influenced by outdoor humidity [41]. Humidity's relationship with air pollutants displays more variability; although it correlates moderately with CO and NO_2, its associations with PM ($PM_{2.5}$ and PM_{10}) are consistently weak [39, 89]. Human occupancy demonstrates a positive relationship with humidity levels, possibly due to respiration and physical activity [59].

VOC demonstrate strong relationships on both indoor environmental conditions and human activity. The air change rate (ACR) and air speed seem to have a considerable impact on VOC concentrations [18, 97]. ACR can also be significantly impacted by human behaviour (e.g. opening windows and doors) [18]. This could potentially be incorporated into a future model directly to potentially improve accuracy. Humidity and temperature have consistent relationships with VOCs [18, 97]. VOCs also correlate with $PM_{2.5}$ and CO_2 [72, 97].

NO_2 similar to other factors, has a strong correlation with its outdoor counterpart [41, 72]. The gas exhibits moderate to strong positive associations with NO, CO_2 and $PM_{2.5}$ [89]. Temperature demonstrates variable relationships with NO_2, showing both positive and negative correlations depending on context, whereas humidity consistently displays an inverse relationship [39, 89].

CO has a strong correlation with humidity, showing moderate positive associations across studies [89, 99]. The gas also shows moderate correlations with PM ($PM_{2.5}$ and PM_{10}), CO_2 and NO_2 [99, 100]. Although temperature displays weaker relationships with CO, its positive correlation persists across multiple studies [89, 99].

O_3 has a limited supply of evidence in the papers reviewed. It is shown that the h of day, day of week, season and temperature are the strong predictors of indoor O_3 [72]. Since ground-level O_3 is a product of reactions between sunlight and pollutants such as NO_2 and VOCs, it may also be the case that these features might be worth incorporating into O_3 predictive models.

The reviewed studies highlight several key patterns in IAQ feature relevance. First, outdoor pollutant levels strongly influence indoor concentrations for $PM_{2.5}$, PM_{10} and NO_2 [39, 41, 72], emphasising the need to account for external environmental conditions. Second, human activity (e.g., occupancy) is a critical driver for CO_2 and VOCs [62, 97], whereas temperature and humidity exhibit complex, context-dependent relationships with other pollutants [89, 98]. Finally, temporal factors (time of day, season) and ventilation dynamics further modulate these interactions [18, 72].

4.2.3.1 | Methods for Feature Selection. Several approaches exist for identifying relevant features in IAQ prediction models. Feature selection methods can generally be categorised under three main types: filter, wrapper and embedded. Filter methods are based off of the data's characteristics using some metric. For example, correlation analysis is commonly used to quantify linear relationships between variables, allowing authors to minimise training time and memory usage while improving model performance [37, 38, 97]. Embedded methods take part during the learning or model building. ML techniques like RF can evaluate feature importance by measuring each variable's contribution to predictive accuracy. Wrapper methods use the same model repeatedly to determine the best features. This includes trial-and-error testing, where models are built with different feature combinations, providing empirical evidence of each variable's impact but can be time consuming if there are a lot of features. Additionally, domain knowledge from environmental science and building engineering can guide informed feature selection based on established pollutant behaviours and indoor air dynamics. These methods are often used in combination to optimise model performance.

4.3 | Research Question 2: What ML Methods of IAQ Forecasting Are Well-Researched?

Selecting an appropriate ML method is crucial for the success of IAQ prediction. The choice of method can have a significant impact on the outcome of the prediction task, as different methods have different strengths and weaknesses. In this section, we explore the most commonly used ML methods in IAQ prediction, their strengths and weaknesses,and how they compare with each other.

4.3.1 | Model Architectures

Multilayer perceptrons (MLP) are a feedforward neural network architecture that consists of layers of nodes with each layer being fully connected to the next [65]. Layers consist of input, hidden and output layers each with several nodes. The output of an MLP is the weighted sum of the inputs and a bias processed through an activation function [93, 98]. Key influencers in an MLPs performance are the number of hidden layers and the number of nodes within these layers as well as the activation function used [13]. In the literature MLP has been found to perform better than SVM at both $PM_{2.5}$ and CO_2 prediction [13, 98]. However, in studies that compare MLP to LSTM, MLP is outperformed by LSTM showing that time-series based methods are more successful in IAQ prediction because of the temporal nature of the data [93]. It has also been found that MLPs perform worse than RF while consuming more computational resources in terms of memory and time to train [65, 93].

Back propagation neural networks (BPNN) are an improvement on standard feedforward neural networks which use a back-propagation algorithm to reduce error [10]. The back-propagation algorithm works by adjusting the weights of layers based on the error of the output using gradient descent [7]. BPNN has been found to have consistently performed poorly when predicting low levels of values, performing worse than SVMs and LSTM models [7].

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Convolutional neural networks (CNN) are a type of neural network that commonly used in image processing tasks; one-dimensional CNNs can be applied to the variables for IAQ prediction [101]. However, these models do not consider the long-term temporal dependencies between variables. CNNs are comprised of input layers, convolutional layers, pooling layers and output layers [16]. The convolutional and pooling layers work to reduce unneeded information and extract features from the data [89]. CNNs are computationally less expensive than some other neural network methods, providing a good advantage in situations where processing power is sparse [16]. CNNs are commonly used in hybrid models where they are used to extract features from the data before being fed into another model such as an LSTM or GRU [89, 101].

Recurrent neural networks (RNN) are a type of neural network that is designed to process sequential data, such as the time series data present in IAQ [16]. At the core of an RNN is the 'hidden cell' which allows them to 'memorise' past information, enabling them to learn from prior inputs and improve predictions [102]. In an RNN, each node computes an output using the previous hidden state and the current input; the output and new hidden state are then fed to the next node in the sequence [102, 103]. This technique allows RNNs to predict the next values in a sequence, taking into consideration how the previous values in the sequence influence the current value. RNNs have a crucial flaw known as the vanishing and exploding gradient problem. This happens because of the weighting matrix being multiplied against previous values repeatedly, which can lead to the value becoming exceedingly small or large causing high computational costs and poor performance in longer datasets [16].

LSTMs are a development of RNNs that address the vanishing and exploding gradient problem. They do this by introducing a memory cell that is comprised of three gates: an input gate, a forget gate and an output gate. The input gate controls how much new information is added to the memory cell, the forget gate is responsible for how much data should be removed from the memory cell and the output gate controls what information is output from the memory cell. This allows LSTMs to selectively remember or forget information, retaining relevant patterns only. They have demonstrated an excellent ability to learn long-term patterns in data, making them very well suited to IAQ prediction tasks [31, 32, 104]. As such, LSTMs are the most commonly used method in the papers reviewed with 23 papers using them. Limitations of LSTMs are that they have a high memory cost and computational time; for example they can take 20–30 times longer than an SVM or twice as long as a BPNN [7]. As well as standard LSTMs there are also bi-directional LSTMs (Bi-LSTM) which process the sequence both forwards and backwards, allowing the model to learn from both past and future information. Bi-LSTM can perform better than standard LSTM in some tasks, but it is not always a clear winner [101].

GRUs simplify the LSTM architecture by combining the input and forget gates into a single update gate, while also merging the cell state and hidden state into a single state. GRUs are generally faster to train while performing comparably to LSTMs in many tasks. Although GRUs are very capable, LSTMs are still often better suited to tasks that require long-term memory. The performance of GRU is frequently comparable to that of LSTM, with multiple papers showing similar scores across multiple different air quality prediction tasks [8, 105]. Additionally, [105] shows that GRU computes faster than LSTM whereas [24] showcases the lower memory footprint of GRU. This could potentially make GRU a better choice for IAQ prediction in cases where computational resources are limited. Similar to LSTM, GRU also has a bi-directional version (Bi-GRU) which processes the sequence both forwards and backwards, allowing the model to learn from both past and future information.

A recent study in an educational building further supports the suitability of GRU and LSTM models for IAQ-related prediction and control. Godasiaei et al. applied RNN, LSTM, GRU, and CNN models to classroom sensor data, including CO_2, $PM_{2.5}$, PM_{10}, temperature, HCHO, TVOC, rain sensor data and window-control variables. Their results showed that GRU achieved the strongest predictive performance, followed by LSTM, suggesting that the architectures are particularly effective for real-time IAQ optimisation and ventilation-control decisions in educational environments [106].

Transformers are a relatively new architecture that was initially developed in 2017 for natural language processing tasks. Transformers use multihead self-attention to process data from multiple subspaces. This allows the model to learn relationships between different parts of the data, regardless of their position in the sequence [107]. In other words, transformers can learn both long-term and short-term dependencies in the data. This makes transformers a potentially very powerful tool for IAQ prediction. In the papers reviewed, transformers were used in five papers [8, 23, 105, 108, 109]. The early findings indicate that transformers can outperform LSTM and GRU in $PM_{2.5}$ prediction, with transformer models performing much better in regard to predicting unhealthy levels of $PM_{2.5}$ [23]. An unfortunate limitation of transformers is that they have a high memory cost compared with LSTM and GRU [105]. Several papers highlight a development on transformers called Informer. Informer has been shown to excel at predicting long-term forecasts, performing notably better than standard transformers, GRU and LSTM [8, 108].

4.3.1.1 | Other ML Methods. RF is a supervised ensemble method for regression and classification, using multiple decision trees trained on random subsets of data and features [14, 98]. Predictions are averaged across trees, making RF less prone to overfitting and more generalisable. Key hyperparameters are the number of trees and features per tree [14]. RF is fast, handles mixed data types and missing values, and is widely used, appearing in eight reviewed papers with training $R^2 > 90\%$ and testing $R^2 > 80\%$, often outperforming ANN, RNN, MLR, SVM, LR, AdaBoost, MLP and Gradient Boost [14, 38, 98]. Limitations include low interpretability, bias towards categorical variables with many levels, and weaker performance than LSTM for complex temporal patterns [14, 72].

SVM maps inputs to a higher-dimensional space to find a separating hyperplane, useful for classification and regression (SVR) [7, 110]. It penalises predictions far from targets and handles high-dimensional, non-linear data [10, 98]. SVM is widely used but results are mixed: it can outperform LSTM for some variables (humidity, VOCs and temperature) but

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underperform MLP for others, possibly due to extreme values [13, 26, 98]. SVM offers lower computational cost compared with LSTM, making it attractive when resources are limited [7].

Hybrid models are models that combine multiple different methods to create a better performing one. The aim is to mitigate the weaknesses of a model by utilising another's strengths. For instance, models of the RNN family require a complete sequence of data to make a prediction, meaning that missing values must be dealt with. To address this, some papers have used other techniques such as RF to predict the missing values before feeding the data into the RNN model [89]. RF has also been used to determine feature importance before feeding the data into a LSTM model, allowing for the model to focus on only relevant features [61]. CNNs have also been used to good effect, reducing dimensionality and extracting features from the data before passing the result to another model such as LSTM or GRU [89, 101].

GANs consist of two competing neural networks: a generator that produces synthetic samples and a discriminator that distinguishes real from fake data. Through this adversarial process, GANs can learn to generate realistic data that mimic the statistical properties of the original dataset. In air quality forecasting, they have been applied to address challenges such as missing or imbalanced data. For example, Asaei-Moamam et al. combined GANs with a neural Turing machine to generate synthetic $PM_{2.5}$, PM_{10} and AQI records, significantly improving prediction accuracy and robustness compared with baseline deep learning models [87]. Other work has highlighted GAN-based augmentation as a promising approach to enrich sparse datasets and improve reliability in time-series air quality prediction [88]. More recently, Leone et al. applied a conditional GAN framework to indoor CO_2 forecasting, using LSTM-based generator and discriminator networks to simulate additional training data from a short real-world monitoring period [66]. Their approach incorporated temperature, humidity, CO_2, and physical activity data collected using environmental sensors and a wearable t-shirt, achieving an RMSE of approximately 8 ppm while reducing the amount of real training data required from 10 days to 5 days. This demonstrates the potential of GAN-based methods not only for data augmentation, but also for reducing data-collection requirements in adaptive IAQ forecasting systems. Despite their potential, GANs are less common in IAQ literature, likely due to the difficulty of generating sequential data that preserves temporal dependencies.

4.3.2 | Model Adaptation

Model adaptation refers to the process of adjusting a trained model so that it performs effectively in new but related contexts. This has the potential to be useful in IAQ forecasting, where differences in buildings, sensors, climates and human activities can cause models trained in one setting to perform poorly in another. Rather than retraining from scratch, adaptation methods enable the reuse of existing models while improving generalisation.

There are many methods of model adaptation. Domain adaptation aligns a model trained on a source domain (e.g., one building or sensor type) with a target domain (another building or sensor), using techniques such as feature alignment, adversarial learning or reweighting of target samples [111]. Transfer learning leverages pretrained models and fine-tunes them on limited data, reducing data requirements [112]. Batch or online learning enables continuous updates as new sensor readings arrive, which is useful for addressing sensor drift or changing environmental conditions [113]. Finally, personalised or context-aware adaptation adjusts predictions based on specific factors such as building structure, occupancy or user health needs [88]. Thomas et al. create a system to detect a data drift condition, and then apply batch learning to address it [113]. This method has the advantage of reducing the amount of computationally expensive training that batch learning requires.

4.4 | Research Question 3: How Does Accuracy Fare Over Prediction Time?

4.4.1 | Forecast Horizon Length

Long-term forecasting of IAQ forecasting introduces a challenge; prediction accuracy consistently degrades as the forecast horizon extends. This occurs because of error accumulation in time-series models and increasing uncertainty in environmental variables. Models excelling at temporal dependency learning demonstrate marked advantages for extended horizons. LSTM, GRU and transformer architectures consistently outperform nontemporal models beyond 30-min forecasts because of their ability to retain long-range dependencies through specialised memory mechanisms [8, 93, 114]. In contrast, nontime-series ML methods like RF and SVM exhibit a larger performance decay at longer horizons, as they lack inherent sequential modelling capabilities [93, 98].

This is shown in the following studies:

- Yang et al. forecast CO_2 concentrations at 5, 10, 15, 20, 25 and 30-min intervals [115]. Evaluation metrics indicate a consistent decline in accuracy over longer time horizons, with RMSE and MAE scores increasing substantially as the prediction window extends, reducing the practical utility for health-related applications over time.
- Taheri and Razban predict CO_2 levels at 1-h, 6-h and 24-h intervals [98]. Performance degrades across all tested models (SVM, RF, LR, AdaBoost, MLP and GB) as the forecasting horizon increases. Among these, the MLP model achieves the best results at all three time intervals.
- Lee et al. evaluate bioaerosol concentration predictions from 10 to 60 min, comparing LSTM against linear regression, lasso regression, RF, XGBoost, MLP and RNN in office and mall environments [93]. The LSTM demonstrates strong performance in learning long-term dependencies, whereas linear and lasso regression models exhibit significant performance degradation over extended forecasting periods.

The consequence of horizon-dependent performance is that short-term forecasts enable proactive interventions like ventilation activation before health-critical thresholds are reached, whereas

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longer predictions become progressively less reliable for health interventions. This underscores the necessity for multihorizon evaluation protocols. Future research should ensure to have a robust evaluation protocol that presents the model's performance at varying forecasting horizons to determine the ideal window.

4.4.2 | Temporal Window Length

Equally important to horizon selection is the length of the temporal input window used for prediction. The input window defines how much historical data is considered when forecasting. Short windows risk missing long-term dependencies such as daily or weekly cycles, whereas very long windows may include redundant information, add noise and increase computational cost [98].

In the papers reviewed, the chosen window length is often unexplained or chosen based on domain knowledge (e.g., 24 h to capture daily cycles), another option is to test many configurations and compare the results [68]. Future research should report both forecast horizon and input window length, and systematically evaluate their effects on prediction accuracy.

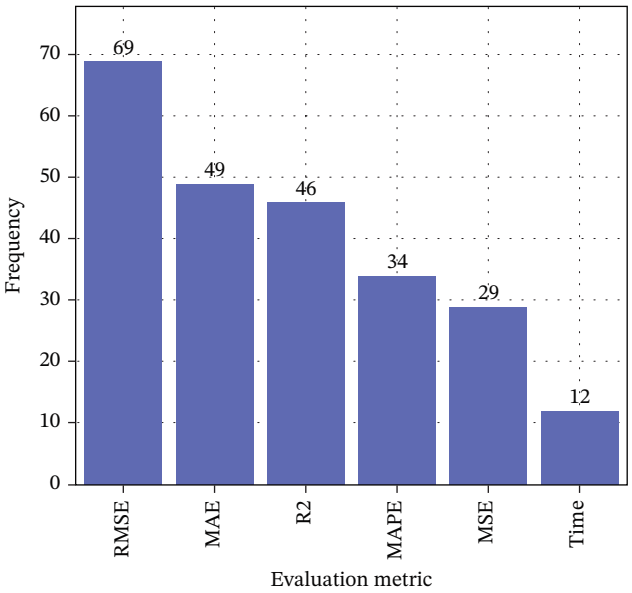


FIGURE 6 | Frequency of the most commonly used evaluation metrics (present in five or more papers).

4.5 | Research Question 4: What Evaluation Metrics Are Suitable for IAQ Forecasting?

An effective evaluation protocol is essential for assessing the performance of IAQ ML forecasting models. An effective evaluation protocol should incorporate a diverse set of metrics that capture different aspects of model performance and should allow for easy comparison between models.

Evaluation of ML models typically makes use of multiple metrics, as no single metric can capture all aspects of a model's performance. Common evaluation metrics include Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R-squared (R^2), Mean Absolute Percentage Error (MAPE), mean squared error (MSE) and computation time. The frequency of these metrics can be seen in Figure 6 and a summary of each in Tables 4 and 5.

As well as following the standard practise for evaluating ML models, there are some special considerations that should be taken into account when evaluating IAQ forecasting models.

When predicting IAQ we are most concerned with the accuracy of the predictions when air quality is poor, as this is the greatest threat to health. However, since the majority of the time air quality is not in these extreme conditions, we frequently have unbalanced data skewed towards average conditions and our models may not perform well at these extreme conditions. Taking an evaluation metric over the entire model's performance could conceal bad performance in these conditions. To address this, authors could evaluate the performance of the model over different bins of the target variable, or they could use an evaluation metric that places emphasis on the model's performance in these extreme conditions.

For example, some papers employ the use of a 'near-the-threshold assessment' whereby three metrics are introduced: critical success index (CSI), false alarm rate (FAR) and probability of detection (POD) [23, 116, 117]. The method works by either categorising the predicted and actual values as not a risk or as a risk to health based on a threshold value. After doing this hits (correctly predicted risks), misses (incorrectly predicted safe) and false alarms (incorrectly predicted risks) are calculated. These values can then be used in the CSI, FAR and POD metrics giving some indication about how well a model could indicate to a user when the air quality is at a level that could be harmful to their health.

$$CSI = \frac{H}{H+M+FA}, \quad FAR = \frac{FA}{H+FA}, \quad POD = \frac{H}{H+M}.$$

Where H, M and FA refer to:

- H (hits): The number of correctly predicted events.
- M (misses): The number of actual events that were not predicted.
- FA (false alarms): The number of predicted events that did not actually occur.

Additionally, [23] also introduce Prediction Interval Coverage Probability (PICP) and Winkler Score (WS) as evaluation metrics. These metrics aim to help evaluate the uncertainty of the model's predictions, which could be particularly useful in IAQ prediction as it could be a good tool for providing users with an indication of how much trust they should place in a prediction. The metrics are defined as follows:

- PICP: The proportion of actual values y_i that fall within the predicted interval $[L_i, U_i]$. A higher PICP indicates better coverage.

$$\text{PICP} = \frac{1}{N}\sum_{i=1}^{N} 1\left(y_i \in [L_i, U_i]\right).$$

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TABLE 4 | Summary of the most commonly used evaluation metrics.

Evaluation Metric	Formula	Description	Advantages	Count
Mean squared error (mse)	$\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \wedge y_i\right)^2$	The average squared difference between predicted and truth values. Measured in units squared.	No particular benefit over other options.	29
Root Mean Squared Error (RMSE)	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \wedge y_i\right)^2}$	RMSE shows the root of the average squared error in the units of the feature.	Easily interpretable because it uses the same units as the target variable; punishes larger errors more severely.	69
Mean Absolute Error (MAE)	$\frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$	The mean error of the results in the units of the target variable.	Easily interpretable because it uses the same units as the target variable.	49
Mean Absolute Percentage Error (MAPE)	$\frac{1}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert \times 100$	The mean error presented as a percentage.	Good for comparing models because it is independent of the scale or unit of the target variable; better than RMSE in the event of outliers.	34
R^2	$1 - \frac{\sum_{i=1}^{n}\left(y_i - \wedge\hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}_i\right)^2}$	The proportion of variance in the target variable represented as a percentage value.	Good for comparing models because it is independent of the scale or unit of the target variable.	46

- WS accompanies PICP by penalising wide intervals and observations outside of the interval. Without this, a PICP interval could encompass all values and seemingly report a perfect score. A good WS score is low.

$$\mathrm{WS} = \begin{cases} (U-L) + \frac{2}{\alpha}(L-y) & \text{if } y < L, \\ (U-L) & \text{if } y \in [L, U], \\ (U-L) + \frac{2}{\alpha}(y-U) & \text{if } y > U. \end{cases}$$

Where y, L, U, N and α, refer to:

- y: The actual observed value.
- L: The lower bound of the prediction interval.
- U: The upper bound of the prediction interval.
- N: The total number of observations.
- α: The significance level, where $(1-\alpha)$ is the nominal coverage probability (e.g., 95%).

In addition, special consideration should be taken to display how a model predicts over different forecast horizons. Longer term predictions may be based off the previous predictions in a time series model; the error will accumulate over time. If a model is evaluated over the entire prediction window, the result is an average meaning the stronger short-term predictions and weaker long-term predictions are not represented very well. To address this, papers should evaluate their models at different prediction windows.

5 | Conclusion

5.1 | Discussion

This systematic review provides a comprehensive overview of the current state of research in IAQ forecasting using ML techniques. The findings highlight an increasing global interest in this area, with a growing number of papers published in recent years. The review identifies $PM_{2.5}$ and CO_2 as the most commonly forecasted pollutants likely due to their significant impact on human health.

5.1.1 | Geographical Under-Representation

As discussed in Subsection 4.1, the reviewed literature exhibited substantial geographical limitations, with most studies conducted in China, Western Europe, South Korea and the United States. In contrast, regions such as Africa, South America, the Middle East, South East Asia, Eastern Europe and Russia have received comparatively little attention. Air quality, both indoor and outdoor, is strongly impacted by climate and regional context, and research conducted in one region may not be generalisable to another. Reasons for this limitation are likely both economical and practical. IAQ forecasting requires large datasets of uninterrupted sensor data; in certain regions this may not be feasible because of unreliable energy infrastructure, geopolitical events or an inability to regularly maintain and calibrate the sensors. Meanwhile, developed regions have a greater ability to allocate funding to IAQ research and a greater concentration of research institutions.

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TABLE 5 | Comparative analysis of ML algorithms for indoor air quality forecasting.

Model category	Example models	Performance ranges	Computational requirements	Strengths	Limitations	References
Neural networks	MLP, 1D-CNN	R2: 0.83–0.99; RMSE: 0.21–21.35; MAE: 0.68–19.46	Moderate; faster than RNNs; GPU optional	Short-term forecasts	Weaker temporal modelling; outperformed by RNNs/transformers	[10, 16, 34, 110, 118]
Recurrent neural networks	LSTM, Bi-LSTM, GRU, Bi-GRU, RNN	R2: 0.83–0.996; RMSE: 8.95–30.04; MAE: 0.68–19.46	High; long training; GPU essential	Long-term dependencies and forecasting; multipollutant forecasting	Computationally expensive; overfitting risk	[22, 24, 35, 58, 97, 98]
Transformer models	Informer, GA-PT-Trans, DSSRNN	R2: 0.89–0.97; RMSE: 5.82–64.13; MAE: 0.27–48.15	Very high; GPU essential	Long-term dependencies and forecasting; complex feature interactions	Huge data requirement; difficult tuning	[8, 23, 105, 108]
Hybrid models	CNN-LSTM, RF-CNN-GRU, FL-CNN-LSTM, DSFA-LSTM-CBAM, TCAN-LSTM	R2: 0.84–0.95; RMSE: 5.82–27.17; MAE: 6.35–20.09	High; feature extraction overhead	Spatial-temporal modelling; multimodality	Complex setup; difficult tuning	[46, 70, 89, 101, 119, 120]
Other ML	RF, SVM, XGBoost, LightGBM, SARIMAX, GradientBoosting	R2: 0.45–0.97; RMSE: 0.21–21.35; MAE: 0.01–17.29	Low–moderate; fast; handles missing data	Simple relationships; fast deployment	Weak temporal learning	[14, 52, 62, 98, 121, 122]
Optimisation algorithms	GA, PSO, BO	N/A	High; tuning intensive	Model optimisation; imbalance handling	Complex; expertise required	[37, 115, 123, 124]

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Similarly, there is a disparity in the types of rooms researched. The overwhelming majority of studies reviewed took place in either an educational, residential or office setting. These are ideal locations to study as they are not too hard to access and monitor, and they are locations where people spend the majority of their lives. However, this does leave other types of environment under-explored; in particular shops, industrial sites, hospitals and restaurants are under-represented despite having interesting dynamics and a high potential for impact.

5.1.2 | Behavioural Data Integration

One of the most notable gaps is the limited incorporation of human behavioural data into prediction models. Despite strong evidence that occupant behaviour significantly influences IAQ, only a minority of reviewed studies integrate such variables. When they do, improvements in model accuracy are commonly observed, especially in the presence of time-series or contextual features like window or door states. This gap is likely related to the practical challenges of collecting behavioural data, including sensor costs, privacy concerns and reliance on participant self-reporting. Certain human-centred data may also be difficult to integrate into a time-series model. For instance, a diary taken by a participant will not have data entries that match the sample rate or the modality of sensor readings. However, the continued omission of this data limits the real-world applicability of many IAQ forecasting models.

5.1.3 | Trends in IAQ Forecasting Methods

Within the review, deep learning techniques are overwhelmingly the most popular category of architecture with LSTMs being the most common for IAQ forecasting. Following LSTM, we found a significant number of papers utilising GRUs, RNNs and Bi-LSTMs. This dominance may be related to the maturity of LSTM and other RNN architectures, with ML frameworks commonly having accessible implementations and therefore reducing the barrier to adoption compared with newer architectures. Although these methods are effective they typically come with larger computational costs and longer training times compared with simpler ML methods. Transformer models are gradually becoming an area of interest, with a small number of papers investigating their use in IAQ forecasting. These models have shown promising results in the initial studies and other domains, but more research is needed to fully understand their potential in IAQ forecasting.

Alhough the usage of deep learning models typically leads to higher predictive accuracy, they are characterised by limited interpretability. The extent of this problem is debatable; however, it may result in apprehension and a slowed adoption in critical environments where transparency and trust are important, such as healthcare facilities and schools. Future research may benefit from integrating model interpretability and expandability techniques, such as feature attribution or attention-based visualisation methods, which can provide some insight into model prediction. Hybrid models, where a deep learning model is paired with a simpler, more table model, may also offer a balance between predictive performance and accuracy.

5.1.4 | Implications of Current Evaluation Practices

Additionally, although standard metrics, such as RMSE and MAE are widely used, few studies adopt metrics that reflect real-world impact, such as threshold-based or uncertainty-based evaluations. This presents a potential disconnect between academic metrics and occupant health outcomes or actionable insights in real-world deployments. As a result, models that perform well under conventional metrics may still fail to provide timely or actionable warnings during high-pollution events, which are the periods of greatest risk. This is particularly important for vulnerable populations, where exposure to high pollution levels can lead to adverse health events, including hospitalisation.

Together these findings indicate that although IAQ forecasting research has made substantial progress, there are still important limitations to be addressed, particularly regarding behavioural integration, evaluation practices and global representation. This review contributes by systematically revealing current approaches, highlighting methodological limitations and identifying priority areas where future work can improve both predictive performance and health outcomes.

5.2 | Future Work

The findings of this systematic review highlight several key areas for future research IAQ forecasting. Broadly, these can be broken down into data input, models and evaluation metrics.

5.2.1 | Data

Firstly, there is a need for more diverse geographic representation in IAQ forecasting studies. We found that developing nations are severely under-represented despite evidence showing the disproportionate impact of poor air quality on these countries and communities.

In terms of data input, the review finds that very few studies incorporate information about the building into their models. This is a significant gap, as building characteristics such as materials, ventilation systems and occupancy patterns can greatly influence IAQ. Additionally, including human behavioural data in IAQ forecasting models typically improves model performance and is critical for reducing exposure by enabling proactive monitoring and behaviour-informed interventions. Despite this, the integration of behavioural data remains limited, highlighting a clear need for further research. We also find that outdoor air quality has a strong relationship with IAQ and can be used to improve IAQ forecasting models. Future research may benefit from integrating outdoor pollutant forecasts into indoor IAQ models, potentially leveraging existing outdoor air quality forecasting research in tandem with IAQ

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research to accurately forecast generate future outdoor data for use in IAQ forecasting.

Air quality forecasting has an inherent data imbalance problem, where readings typically fall within an acceptable range and only a small percentage of readings are outside of this range (i.e. highly polluted). This can lead to models that perform great on average but fail to accurately predict these critical high-pollution events. A critical step in the future of IAQ forecasting is to address this data imbalance problem, potentially through synthetic data generation or techniques that penalise models for failing to predict these important events.

5.2.2 | Models

ML architectures, such as LSTM and GRU have been thoroughly explored in the literature, showing great results. A potential direction for future research is to explore the use of transformer models in IAQ forecasting. Although there are some studies that have begun to investigate this, there is still strong potential for further exploration. Transformer models have shown great promise in other domains, such as natural language processing and computer vision, and their application in IAQ forecasting could lead to improvements in prediction accuracy and model robustness.

Additionally, there is an absence of papers that explore relearning and model adaptation to changing conditions. This is a significant gap, as IAQ can vary greatly over time and across different environments. Future research could explore the use of online learning or transfer learning techniques to adapt models to new conditions, potentially improving their accuracy and robustness.

Another important research gap concerns the selection of temporal input windows. There is limited systematic evaluation of how different window lengths affect predictive performance across pollutants, building types or seasons. The choice of window length directly influences how much historical information the model can leverage, and inappropriate choices may either omit relevant dynamics or introduce unnecessary noise. Future work should therefore explore adaptive or optimised approaches for window selection, potentially integrating automated hyperparameter tuning or attention mechanisms, to ensure that models capture the most informative temporal dependencies for accurate and robust IAQ forecasting.

5.2.3 | Evaluation Metrics

There is a lack of a standardised approach to evaluating IAQ forecasting models. Typically authors will use a subset of the common evaluation metrics (RMSE, R2, MSE, MAE and MAPE) making it difficult to compare against another paper using different metrics. Additionally, the need for context-aware evaluation metrics is evident, as current metrics do not adequately reflect the real-world impact of IAQ forecasting models. Establishing a standardised set of evaluation metrics that consider the performance of models at predicting different severities of pollution as well as performance over time could greatly enhance the applicability and comparability of papers. Accurate prediction of high-concentration pollution events is critical for minimising exposure among vulnerable populations, such as individuals with respiratory illnesses, and may reduce the risk of adverse health events, hospitalisations and lost work.

5.3 | Limitations

This systematic review has several limitations. Firstly, the search was limited to papers published in English, which may have excluded relevant research published in other languages. Secondly, because the studies being concentrated in China, South Korea, Europe and the United States, the findings may not be entirely generalisable to other regions where sensor availability, climate, building materials and cultural practices may differ significantly. In addition, this review focused on comparing methods, datasets and evaluation practices rather than statistically combining performance results. Because of differences in pollutants, forecasting horizons, datasets, and evaluation metrics across studies, direct numerical comparison of model performance was not appropriate. Therefore, findings are presented through descriptive comparison and discussion. Finally, although themes and trends were identified through a structured and consistent data extraction process, the synthesis involved an element of judgement when grouping studies and identifying patterns.

Author Contributions

Lewis Hall Rudford: conceptualization, data curation, formal analysis, investigation, methodology, project administration, software, visualisation, writing—original draft. **Sara Sharifzadeh:** conceptualization, funding acquisition, project administration, resources, supervision, validation, writing—review and editing. **Nicholas Micallef:** conceptualization, funding acquisition, project administration, resources, supervision, writing—review and editing; **Paul Lewis:** conceptualization, funding acquisition, resources, writing—review and editing. **Philip Webb:** conceptualization, funding acquisition, resources, writing—review and editing; **Daniele Doneddu:** conceptualization, funding acquisition, project administration, resources, supervision, writing—review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Supporting Information.** Data S1: Prisma 2020 checklist.

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Appendix A: Search strategy and database queries

TABLE A1 | Searches performed in the selected databases.

Database	Query	Number
ACM	*Title:(("indoor air quality" OR "indoor environment quality") AND (predict * OR forecast *) OR Abstract:(("indoor air quality" OR "indoor environment quality") AND (predict * OR forecast *) AND ("model" OR "algorithm" OR "develop * ")) OR Keyword:(("indoor air quality" OR "indoor environment quality") AND (predict * OR forecast *) AND ("model" OR "algorithm" or "develop * "))*	38
IEEE Xplore	*(("Abstract":"indoor air quality" OR "Abstract":"indoor environment quality") AND ("Abstract":"predict * " OR "Abstract":"forecast * ") AND ("Abstract":"model" OR "Abstract":"algorithm" OR "Abstract":"develop * ")) OR (("Document Title":"indoor air quality" OR "Document Title":"indoor environment quality") AND ("Document Title":"predict*" OR "Document Title":"forecast*")) OR (("Author Keywords":"indoor air quality" OR "Author Keywords":"indoor environment quality") AND ("Author Keywords":"predict*" OR "Author Keywords":"forecast*") AND ("Author Keywords":"model" OR "Author Keywords":"algorithm" OR "Abstract":"develop*"))*	80
ProQuest	*Title((("indoor air quality" OR "indoor environment quality") AND (predict * OR forecast *))) OR abstract((("indoor air quality" OR "indoor environment quality") AND (predict * OR forecast *)) AND ("model" OR "algorithm" OR develop *))*	1073
PubMed	*(((indoor air quality[Title/Abstract]) OR (indoor environment quality[Title/Abstract])) AND ((predict * [Title/Abstract]) OR (forecast * [Title/Abstract])) AND ((model[Title/Abstract]) OR (algorithm[Title/Abstract]) OR (develop * [Title/Abstract])))*	241
Science Direct	*("indoor air quality" OR "indoor environment quality") AND (prediction OR forecast) AND (model OR algorithm OR develop)*	335
Scopus	*(TITLE-ABS-KEY ("indoor air quality") OR TITLE-ABS-KEY ("indoor environment quality") AND TITLE-ABS-KEY (predict *) OR TITLE-ABS-KEY (forecast *) AND KEY (model) OR TITLE-ABS-KEY (algorithm) OR TITLE-ABS-KEY (develop *))*	1479
Web of Science	*((TI = (("indoor air quality" OR "indoor environment quality") AND (predict OR forecast))) OR AB = (("indoor air quality" OR "indoor environment quality") AND (predict OR forecast) AND ("model" OR "algorithm" OR "develop * "))) OR KP = (("indoor air quality" OR "indoor environment quality") AND (predict * OR forecast *) AND ("model" OR "algorithm" OR "develop * "))*	943
		Total: 4,189

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